

FEBRIAN FITRYANIK SUSANTA

Date of birth: Madiun-Indonesia, February 16th, 1995

Nationality: Indonesian

Affiliation: Department of Geomatics, National Cheng Kung University,

Department of Geodetic Engineering, Universitas Gadjah Mada, Yogyakarta, Indonesia

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EDUCATION

Doctoral's Degree (February 19, 2024 to now (ongoing)) :

- Ph.D. student at the Department of Geomatics, National Cheng Kung University, Taiwan
- 18 credits completed | Weighted average score: 93.4/100

Master's Degree (August 8, 2016 to June 8, 2018) :

- M.Eng in Geomatic Engineering, Universitas Gadjah Mada (UGM), Indonesia
- GPA: 3.93 (4.0 Scale) | Number of Courses: 16 | Number of Credits: 42
- Thesis titled (in Indonesian): "Network Analysis for Modeling Bus Public Transport Systems using Weight Parameters in Kulon Progo Regency"

Bachelor's degree (September 1, 2013 to July 21, 2017) :

- S.T. (Engineer) in Geodetic Engineering, Universitas Gadjah Mada, Indonesia
- GPA: 3.87 (4.0 Scale) | Number of Courses: 61 | Number of Credits: 151
- Thesis titled (in Indonesian): "Visual Analytics of Spatio Temporal Data for Railway Network using Interactive Online Map"

Graduate Selected Courses in Geodetic and Geomatic Engineering:

Geodetic Surveying, Remote Sensing and Image Interpretation, Database Systems, Computer Graphics, Geographical Information Systems, Photogrammetry, Geospatial Analysis, Spatial Statistics, Applied GIS, Cartography, Web GIS, Spasial Data Infrastructure

EXPERIENCE

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|----------------------------|---|
| Nov. 2020 to now (ongoing) | <u>Assistance Professor at the Department of Geodetic Engineering, UGM</u>
Designed and delivered undergraduate and graduate-level courses; Supervised theses and capstone projects; Published peer-reviewed journal articles |
| 2016 to 2024 | <u>Trainer in Basic Geographic Information Systems (GIS) and WebGIS at Frasta Education Center, Center for Transportation and Logistics Studies UGM, and as an incidental trainer.</u>
Taught core courses; Initiated project-based learning modules; Organized participants workshops |
| Nov. 2018 to Oct. 2020 | <u>Lecturer at the Department of Geodetic Engineering, UGM</u>
Taught core courses; Initiated project-based learning modules; Organized student workshops on GIS tools |

Nov. 2018 to	<u>Researcher of Center for Spatial Data Infrastructure Development, UGM</u>
Jan. 2024	Contributed to national and regional research projects; Co-authored technical reports and presented findings; Developed prototype spatial data platforms
Before Nov. 2018	<u>Teaching and research assistant, Department of Geodetic Engineering Universitas Gadjah Mada, Indonesia</u>
	Supported faculty in delivering lectures and lab sessions and participated in faculty-led research

SELECTED PUBLICATIONS

- Putra, I. G. B., Kuo, P. F., Susanta, F. F., Tedjo, B. H., & Lord, D. (2026). GeoShapley-based interpretation of older adult pedestrian fatal vs injury crash frequency in dense urban environments. *Accident Analysis & Prevention*, 230, 108450.
- Tedjo, B. H., Kuo, P. F., **Susanta, F. F.**, & Putra, I. G. B. (2026). Identifying environmental factors related to motorcyclist crash rates: variable selection using spatial Random Forest with network distance and barriers. *Accident Analysis & Prevention*, 226, 108362.
- Susilo, H., Rassarandi, F.D., **Susanta, F.F.**, Putra, Komariah, S. 2020. Exploration of Various Visual Display Methods for Geovisualization Purposes. *Prokons: Jurnal Teknik Sipil*, 14(1). <http://dx.doi.org/10.33795/prokons.v14i1.293>.
- Marwan, Wirandha, F.S., Nizzamuddin, and **Susanta, F.F.**, 2020. Perbandingan Ketelitian Foto Udara UAV (Unmanned Aerial Vehicle) dan GNSS (Global Navigation Satellite System) pada Pemetaan Gampong Lambarih, Aceh Besar, Aceh. *Journal of Aceh Physics Society*, 9(3). <https://doi.org/10.24815/jacps.v9i3.17151>.
- Susanta, F.F.**, Aditya, T., 2020. Visualisasi Pemodelan Hasil Analisis Jaringan Angkutan Umum di Kabupaten Kulon Progo. *GEOMATIKA: Jurnal Badan Informasi Geospasial*, 26(1). <http://dx.doi.org/10.24895/JIG.2020.26-1.1085>.
- Aditya, T., Laksono, D., **Susanta, F.F.**, Istarno, I., Diyono, D. and Ariyanto, D., 2020. Visualization of 3D Survey Data for Strata Titles. *ISPRS International Journal of Geo-Information*, 9(5), p.310. <https://doi.org/10.3390/ijgi9050310>.
- Susanta, F.F.**, Pratama, C., Aditya, T., Khomaini, A.F. and Abdillah, H.W.K., 2019. Geovisual Analytics of Spatio-Temporal Earthquake Data in Indonesia. *JGISE: Journal of Geospatial Information Science and Engineering*, 2(2). <https://doi.org/10.22146/jgise.51131>.
- Pratama, C., **Susanta, F.F.**, Ilahi, R., Khomaini, A.F. and Abdillah, H.W.K., 2019. Coseismic Displacement Accumulation Between 1996 and 2019 Using A Global Empirical Law on Indonesia Continuously Operating Reference Station (InaCORS). *JGISE: Journal of Geospatial Information Science and Engineering*, 2(2). <https://doi.org/10.22146/jgise.51130>.
- Rassarandi, F.D., Susilo, H., Kariyono, K., Fajri, K. and **Susanta, F.F.**, 2019. The Evolution of Geospatial Aspects in the Law on the Establishment of New Regions (Autonomy) Pre and Post Broad Autonomy in Indonesia. *Journal of Applied Geospatial Information*, 3(2), pp.223-227. <https://doi.org/10.30871/jagi.v3i2.1309>.
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SELECTED CONFERENCES

- Susanta, F. F.**, Kuo, P.-F., & Putra, I. G. B. (*accepted for presentation*). Modeling dynamic walkability to support time-based route planning for older adults. In *Proceedings of the XXV ISPRS Congress (TC IV: Spatial Information Science)*, Toronto, Canada.
- Ku, T. Y., Susanta, F. F., Kuo, P. F., Shen, C. W., Lai, C. J., Chen, N. Y. (*accepted for presentation*). Integrating Multi-objective Optimization and Spatial Analysis for the Site Selection and Improvement of Levee Crossing Facilities. World Conference on transport research: WCTR 2026. Toulouse, France.
- Putra, I. G. B., **Susanta, F. F.**, Tedjo, B. H., & Kuo, P. F. (2025). Interpretable machine learning for crash severity analysis of food delivery motorcyclists. In *Proceedings of the 46th Asian Conference on Remote Sensing (ACRS 2025)*, Makassar, Indonesia.
- Ku, T. Y., Susanta, F. F., Kuo, P. F., Shen, C. W., Lai, C. J., Chen, N. Y. (2025). Integrated Multi-Objective Optimization and Spatial Analysis for the Site Selection of Levee Crossing Facilities. 2025 Informs Annual Meeting. Atlanta, Georgia. <https://meetings.informs.org/wordpress/annual/>
- Li, Y. T., Kuo, P. F., Chiang, K. W., Putra, I.G.B., & **Susanta, F. F.** (2025). Utilizing High-Definition Maps and Simulation Software to Enhance Autonomous Vehicle Safety. *The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, 48, 929-935. Dubai.
- Susanta, F. F.**, Kuo, P. F., Putra, I.G.B., and Tedjo, B.H., 2025. Equitable Sidewalks for All: Enhancing Hospital Pedestrian Accessibility Through Data-Driven Assessment. In *The 30th International Symposium on Remote Sensing*. <https://isrs.or.kr/>. Incheon, South Korea.
- 李怡葶、郭佩棻、Febrian Fitryanik Susanta, 2024 提高自動駕駛汽車安全性：利用高精度 地圖和 Carla軟體建置自駕車之操作設計領域. In *TGIS 2024 台灣地理資訊學會年會暨學術研討會*. <https://tgis2024.rchss.sinica.edu.tw/site/default.aspx?sid=1553&lang=cht>
- 宏觀、鄭宛妍、郭佩棻, Febrian Fitryanik Susanta, 2024. 人行空間品質測量與空間分析-以老年行人與外籍看護為例. In *TGIS 2024 台灣地理資訊學會年會暨學術研討會*. <https://tgis2024.rchss.sinica.edu.tw/site/default.aspx?sid=1553&lang=cht>
- Susanta, F. F., Kuo, P. F., & Putra, I.G.B., 2024. A Comparison of the Spatial Accessibility Model for Hierarchical Healthcare Facilities in Yogyakarta City, Indonesia. In *The 29th International Symposium on Remote Sensing*. <https://sites.google.com/view/2024isrs>.
- Susanta, F. F.**, Muryamto, R., & Laksono, D. P. (2024, January). The 15-minute city of mass public transportation in Yogyakarta: case of Trans Jogja bus. In *Eighth Geoinformation Science Symposium 2023: Geoinformation Science for Sustainable Planet (Vol. 12977, pp. 25-32)*. SPIE.
- Rokhmana, C., Susanta, F.F., and Setyanegara, S., 2023. Utilizing The Low-Cost GNSS Module, Drone, and Smartphone Device for Precise Asset Monitoring System. In *South East Asia Workshop on Geodetic Data Sciences, Geoinformatics, and Land Administration (Geo-Land-Sea) 2023*. <https://geodesi.ugm.ac.id/proceeding-geolandsea2023/>
- Rahmawati, A., Yuliandari, S.S., Susanta, F.F., Jaya, I, Kosasih, B. 2022. Development of 3D Web-GIS for Ambon Flood Hazard Mitigation using Mathematical Modeling, Lidar and GIS. In *South East Asian Surveyor Congress 2022*.
- Prasetya, S.R. and Susanta, F.F., 2022. Visualisasi Hasil Analisis Dampak Bencana Gunung Api Sinabung Tahun 2018 – 2020. In *Seminar Nasional Geomatika (pp. 1015-1026)*.
- Yudhanti, G.T, Susanta, F.F., and Pratama, C., 2022. Visualisasi Risiko Bencana Gempa Bumi Menggunakan Webgis untuk Mitigasi Bencana (Studi Kasus: Kota Palu, Sulawesi Tengah). In *Seminar*

- Nasional Geomatika (pp. 1-10).
- Aditya, T., Laksono, D.P., Atunggal, D., **Susanta, F.F.**, Widjajanti, N., Setiawan, M.B., Agam, N. and Wibisono, T., 2021. 3D Modelling, Validation and Visualization of 3D Parcels in First Registration for 3D Cadastre-Indonesia Case. In 7th International FIG Workshop on 3D Cadastres. <https://doi.org/10.4233/uuid:6669b973-e44f-4009-8127-030d4bbf6696>
- Aryani, D.D., SP, D.A. and Susanta, F.F., 2021. Pembuatan Sistem Informasi Kondisi Jalan Pada Ruas Jalan di Kabupaten Indragiri Hulu. In Seminar Nasional Geomatika (pp. 523-534).
- Susanta, F.F., Pratama, C., Khomaini, A.F. and Abdillah, H.W.K., 2019. Pemodelan Variabel Gempa Bumi sebagai Sarana Edukasi untuk Mitigasi Bencana di Indonesia. Forum Ilmiah Tahunan – Ikatan Surveyor Indonesia. Jakarta. <https://drive.google.com/file/d/1ApkerSwFaeALmcY6QbRil2w38hGog7hV/view>.
- Susanta, F.F., 2019, November. Visual analytics for railway network in Java Island. In Sixth Geoinformation Science Symposium (Vol. 11311, p. 1131108). International Society for Optics and Photonics. <https://www.spiedigitallibrary.org/conference-proceedings-of-spie/11311/2548481/Visual-analytics-for>.
- Aditya, T., Laksono, D., Sutanta, H., Izzahudin, N., and Susanta, F., 2018. A Usability Evaluation of A 3D Map Display For Pedestrian Navigation, Int. Arch. Photogramm. Remote Sens. Spatial Inf. Sci., XLII-4/W10, 3-10, <https://doi.org/10.5194/isprs-archives-XLII-4-W10-3-2018>, 2018.
- Susanta, Febrian; Aditya, Trias. 2018 Visualisasi Permintaan Penumpang Angkutan Umum Hasil Analisis Analytical Hierarchy Process (AHP) di Kabupaten Kulon Progo. Simposium Infrastruktur Informasi Geospasial. Yogyakarta

SELECTED BOOKS

- Heliani, S., **Susanta, F.F.**, (2023). Teknologi dan Tata Kelola Produksi Garam: Aspek Geografis: Analisis Potensi Lokasi Produksi Garam. In Ivan, A.A. (Eds.), Teknologi dan Tata Kelola Produksi Garam (pp. 1-25). Pustaka Belajar.
- Pribadi, O.D., Adrianto, D., Amarona, M.Q., Mulyadi, D.S., Abimanyu, A., Dharma, C.S., **Susanta., F.F.**, (2021). Jala Citra I - 2021 Aurora Expedition. Hydro-Oceanography Centre Indonesian Navy.

HONORS AND AWARDS

- 2025 – Humanistic Street Hackathon (College and University Student Competition), Ministry of the Interior, Land Management Agency, Taiwan: Excellent Team (Group Project) for Data Selection, Smart Embankment Crossing: Extending Humanistic Street to the River Embankment.
- 2024 – Veritas et Conscientia Scholarship (VCS) Elite Scholarship, National Cheng Kung University.
- 2017 – Best Graduate, Bachelor of Engineering, Department of Geodetic Engineering, Universitas Gadjah Mada (UGM).
- 2017 – Top 3, Infrastructure and Transportation Category, Tokyo-Tech Innovation Commitment Award.
- 2016 – 1st Place, Community Empowerment for Road Safety, Directorate General of Land Transportation, Ministry of Transportation, Indonesia.
- 2016 – Top 5, Geosat Scientific Writing Competition, Institut Teknologi Sepuluh Nopember (ITS), Surabaya.

SELECTED CERTIFICATES

- LiDAR Winter School, National Cheng Kung University (2026)
- Lecturer Certification, Indonesian Ministry of Education, Culture, Research, and Technology (2023)
- Professional Engineer Certification (Ir.), Institution of Engineers Indonesia (PII / Persatuan Insinyur Indonesia) (2023)
- Public Awareness Program on Geographical Names and Toponyms, Geospatial Information Agency Indonesia (2023)
- Networking Development Training, Universitas Gadjah Mada (2021)
- IELTS Advanced, REAL - IONs International Education (2021)
- MOOC “Spatial Data Science – The New Frontier in Analytics”, ESRI (2020)
- MOOC “The Location Advantage”, ESRI (2020)
- Geographic Information System Training, ESRI (2020)
- IELTS Foundation, REAL - IONs International Education (2020)
- Training of Trainers (ToT) Faculty Advisor for the Student Creativity Program (PKM) (2019)
- MOOC “Going Places with Spatial Analysis”, ESRI (2019)
- MOOC “Cartography”, ESRI (2019)
- Innovative Learning Ecosystem Training, UGM (2019)
- Assistance and Supervision of Regional Spatial Planning Mapping, Geospatial Information Agency Indonesia (2019)
- Workshop on Academic Writing and Publishing, Nature Research Academy (2019)
- Workshop on Academic Writing and Publishing, Enago Academy (2019)

ASSOCIATIONS

- Indonesian Surveyor Association (ISI / Ikatan Surveyor Indonesia) <http://www.isi.or.id/>
- Institution of Engineers Indonesia (PII / Persatuan Insinyur Indonesia) <http://www.pii.or.id/>
- The ISPRS Student Consortium (ISPRS SC) <https://sc.isprs.org/>

RESEARCH, PROFESSIONAL PROJECTS, AND COMMUNITY SERVICE EXPERIENCE

- 2024–Present – Elderly Pedestrian Traffic Safety and Pedestrian Space: A Case Study of Sidewalk Characteristics. Research Assistant. Higher Education Sprout Project, Ministry of Education (NCKU), with additional support from the National Science and Technology Council (NSTC), Taiwan.
- 2023 – Study on Pilot Project for Upgrading 3D Spatial Data (Bogor, Indonesia). Mid-level Researcher. Ministry of Agrarian Affairs and Spatial Planning / National Land Agency, Indonesia.
- 2023 – Community Service in Murtigading Village, Sanden Sub-district, Bantul Regency Based on a Geospatial Approach. Community Service Member. Department of Geodetic Engineering, Universitas Gadjah Mada.
- 2023 – Accessibility Analysis of Public Transportation in Yogyakarta City. Team Leader. Department of Geodetic Engineering, Universitas Gadjah Mada.
- 2022 – Potret 1200: Asset Monitoring System Based on UAV and Internet. Researcher. Kedaireka (Indonesian Sovereignty in Design).
- 2021–2023 – Development of Spatial Information System (<http://geoportal.ppids.ft.ugm.ac.id/sitaru/>). GIS Expert. Land and Urban Planning Agency, Yogyakarta City.

- 2021–2023 – Community Service in Gerbosari Village, Samigaluh Sub-district, Kulon Progo Regency Based on a Geospatial Approach. Community Service Leader. Department of Geodetic Engineering, Universitas Gadjah Mada.
- 2021 – Land Database Development for Implementing 3D Cadastre. Co-Researcher. Ministry of Agrarian Affairs and Spatial Planning / National Land Agency, Indonesia.
- 2021 – Jala Citra I – Aurora Expedition. Researcher. Hydro-Oceanography Centre, Indonesian Navy.
- 2020–2023 – Development of Karang Sari Village Geospatial Information System, Pengasih, Kulon Progo, Yogyakarta (Community Service). Community Service Member. Faculty of Engineering, Universitas Gadjah Mada.
- 2020–2021 – Collaborative Land Boundary Delimitation Using AR/VR Technology Based on Game Engines to Accelerate Complete Systematic Land Registration. Co-Researcher. Higher Education Applied Research, Indonesia.
- 2020 – Consultancy Services for National Spatial Data Infrastructure (NSDI) Review. Research Assistant. Geospatial Information Agency, Indonesia.
- 2020 – Pilot Project of 3D Cadastre (MRT Jakarta, Indonesia). Co-Researcher. Ministry of Agrarian Affairs and Spatial Planning / National Land Agency, Indonesia.
- 2019 – Preparation of Land Administration Information System Planning in Madiun Regency. Team Member. Development Planning Agency at Sub-National Level, Indonesia.
- 2019 – Development of Spatial Information System (geoportal.klatenkab.go.id/sitaru). GIS Expert. Office of Spatial Planning and Public Works, Klaten Regency, Indonesia.
- 2019 – Updating and Development of Android Application Layouts for Spatial Information Systems (<https://sitaru.jogjakota.go.id>). GIS Expert. Land and Urban Planning Agency, Yogyakarta City.
- 2019 – GIS Training for Village Staff in Panjatan Sub-district, Kulon Progo, Yogyakarta (Community Service). Community Service Member. Faculty of Engineering, Universitas Gadjah Mada.
- 2019 – Development of Sidokarto Village Geospatial Information System, Godean, Sleman, Yogyakarta (Community Service). Community Service Member. Department of Geodetic Engineering, Universitas Gadjah Mada.
- 2019 – Visual Analytics of Spatiotemporal Data for Earthquake Potential Mapping in Indonesia. Team Leader. Department of Geodetic Engineering, Universitas Gadjah Mada.
- 2018–2019 – 3D GeoSmartCity. Research Assistant. Higher Education Applied Research, Indonesia.
- 2017 – Development of Geocrowd and Geogaming Applications for Crowdsourcing Peat Fire Monitoring in Pulau Padang, Riau. Research Assistant. Six Capital – Universitas Gadjah Mada.

PERSONAL SKILLS

- Productivity Software: Microsoft Word, Excel, PowerPoint, Google Workspace (Docs, Sheets, Slides)
- Programming Skills: C++, Python, JavaScript, R, Matlab
- Data Analysis & Visualization Tools: Tableau, DataPlotly, Microsoft Power BI, ArcGIS Online Apps
- GIS & Remote Sensing Software: QGIS Desktop, ArcGIS Desktop, ArcGIS Pro, Google Earth Engine, Cloud Compare, Global Mapper
- Database Management: PostgreSQL/PostGIS
- Web Mapping & Spatial Web Development: LeafletJS, D3.js, Carto, OpenLayers, Mapbox

- Geoportal Development: Geonode
 - Languages: Indonesian (Native), English (Proficient)
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Tainan, February 6th, 2026

A handwritten signature in black ink, appearing to be 'Febrian Fitryanik Susanta', written over the printed name.

Febrian Fitryanik Susanta

A small, handwritten signature or mark at the bottom of the page.